



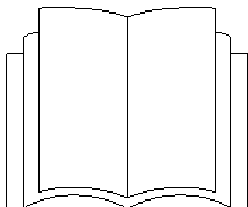
Annex for the original attachment's Instruction Manual



Log Cutter/Splitter

JAPA 300

Product number:	300-700-series loaders:	A21029
	200-series loaders:	A21262



Please read carefully the original attachment's instruction manual before using the equipment, as well as this annex, and follow all instructions.

This annex for the operator's manual includes specific information about the use of the equipment with Avant multi purpose loaders.

Keep the original operator's manual as well as this annex for later reference.

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This manual uses the following warning symbols to indicate important points that must be taken into account to reduce the risk of personal injury or property damage:

	DANGER:
	This safety symbol refers to important safety information in this manual. It warns of an immediate hazard that could cause serious personal injury or property damage. Read the warning text accompanying the symbol carefully and ensure that other operators are also familiar with the warnings, since personal safety is at stake.



Note:
This notice symbol indicates information about the correct operation and maintenance of the equipment.

Failure to observe the instructions accompanying the symbol can lead to equipment breakdown or other property damage.

In addition to the safety instructions included in this manual, you must observe all occupational safety regulations, local laws and other regulations concerning the use of the equipment. Due to continuous product development, some of the details in this manual may differ from your equipment. We reserve the right to change the contents of the manual without notification. This annex for the instruction manual is a translation of the original Finnish version.

1. Foreword

Avant Tecno Oy would like to thank you for your purchase of this attachment for AVANT multi purpose loaders. The log cutter/splitter has been adapted for use with AVANT loaders by the original manufacturer of the equipment.

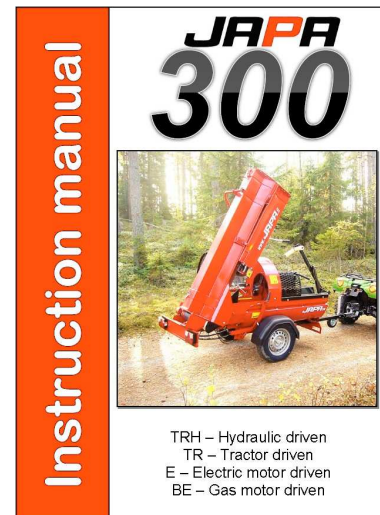
The original manual, provided with the log splitter/cutter, includes the information of safe use of the equipment. This annex for the manual includes important information of mounting and using the equipment with AVANT loaders.

By familiarising yourself with the original operator's manual and the additional instructions provided in this annex, you ascertain your safety and ensure the reliable operation and long service life of the equipment. Ensure that all persons using the loader and the log cutter/splitter have received proper guidance and familiarised themselves with the manuals of the loader and attachment and all safety instructions before using the equipment.

Keep this annex of the manual at hand throughout the service life of the equipment. If this annex for the manual or the original operator's manual is lost or damaged, you can request a new one from your nearest AVANT dealer. If you sell or transfer the equipment, be sure to hand both parts of the manual to the new owner.



This annex for the original instruction manual of the log cutter/splitter machine does not cover all of the instructions needed for safe operation of the equipment. It is not meant to be distributed separately. The original instructions manual, provided with the machine, should be considered as the main source of information about the safety, use and the maintenance of the equipment. This annex includes important information about the use of the equipment specifically with AVANT multi purpose loaders.



2. Designed purpose of use

The JAPA firewood machines are hydraulically operated attachments designed and manufactured to be suitable for use with 200-700 –series AVANT multi purpose loaders. They are intended to be used by one person at a time for cutting firewood and splitting it into halves or quarters. The log cutter/splitter cuts the log with a chain saw blade to the adjusted length, and then the log is split with a hydraulic cylinder, which presses the log against the splitting wedge.

The log cutter/splitter is meant to be attached on the AVANT loader during use. The attachment is powered by the hydraulic outlet of the Avant loader. The log conveyor, shown in the original manual, is an optional extra. Contact your AVANT dealer for further information about its availability.

Familiarise yourself with the instruction manual's instructions regarding service and maintenance. Demanding repair and maintenance operations, such as hydraulic repairs, should be left for professional maintenance. Please contact your AVANT retailer if you have additional questions about the operation or maintenance of the equipment or if you require spare parts or maintenance services.

3. Safety instructions for using the log splitter with Avant loaders

Please bear in mind that safety is the result of several factors. The loader-attachment combination is highly powerful, and improper or careless use or maintenance may cause serious personal injury or property damage. Due to this, you should carefully familiarise yourself with the manuals of both the loader and the attachment before starting operation. Do not use the attachment if you have not completely familiarised yourself with its operation and the related hazards.



- When mounting the log cutter/splitter, ensure that the **locking pins** of the loader's quick attach plate are **in the lower position** and that they have **locked** the attachment **to the loader**.
- The log cutter/splitter is to be used by one operator at a time. Keep others at a safe distance, and do not let others to operate the controls of the loader while using the attachment.
- When transporting the equipment, pay attention to the contours of the terrain as well as other hazards, such as loose rocks or branches and trees reaching the driver's area.
- Always transport the log cutter/splitter as low and close to the machine as possible to keep the centre of gravity low. Keep the telescopic boom retracted.
- Keep hands and feet away from moving parts.
- When detaching the log cutter/splitter, ensure that it is secured against accidental movement.
- Never perform any maintenance or adjustments while the loader engine is running.
- Do not modify the structure of the attachment in a manner that would affect its safety.
- Also read the safety instructions for the loader in the loader manual. Get to know the loader controls and how to drive it safely.



Familiarise yourself with the loader controls in a safe area. Pay particular attention to the safe stopping of the attachment and the machine.



Note that the loader boom moves down and may be tilted even if the engine has been stopped. Never go under or in front of raised boom or attachment.



Note the effects of heavy loads to the the stability and handling characteristics and of the loader, especially when operating with the smaller loader models. Always keep the load close to the ground and keep the telescopic boom retracted. Drive slowly and carefully and avoid steep slopes. Use additional counterweights as needed. Read the operator's manual of the loader.



Remember to wear proper protective clothing:



- The noise level in the driver's seat exceeds 85 dB(A). Wear hearing protection.
- The use of safety boots is recommended while working with the loader.



- Wear protective gloves.

4. Technical specifications

On the original instruction manual are listed the labels and markings, which must be visible on the equipment. Replace the warning labels, if they have become unclear or detached completely. New labels are available via your retailer.

Table 1 –Log cutter/splitter specifications

Product number	A21262	A21029
Log length, min-max:	200 - 500 mm	
Log diameter, min-max:	20 - 300 mm	
Weight:	265 kg	
Minimum operating temperature:	-25°C	
Saw flange:	13"	
Splitting force:	30 kN	
Dimensions, transport configuration: L x W x H:	2780 x 1030 x 2450 mm	
Maximum input of hydraulic energy:	40 l/min, 22,5 MPa (225 bar)	
Suitable AVANT loader:	216, 218, 220	320-750

5. Assembly of the log cutter/splitter

The log cutter/splitter is ready for use as delivered. Attach the attachment and its hydraulic hoses according to the following instructions.

5.1. Attaching the log cutter/splitter to the loader

Mounting the attachment to the loader is quick and easy, but it must be done carefully. If the attachment is not locked to the loader, it may detach from the loader and cause hazardous situations. In the worst case, the attachment may fall on the operator or slide onto the driver along a raised boom. For this reason, the attachment must under no circumstances be raised to a height of over one metre when it has not been locked. To prevent hazardous situations, always follow the attachment mounting instructions shown in the following pages. Also remember the safety instructions in Section 3 and the information in the original instruction manual.

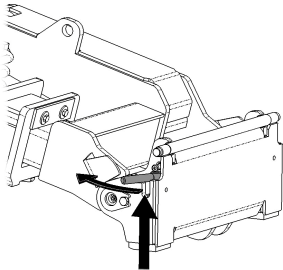
The attachment is mounted to the AVANT loader boom quick attach plate and the counterpart on the attachment. The locking is semi-automatic, but due to the relatively high position of the quick attach plate on the log splitter, and the risk of tipping over of the attachment, the locking pins must be locked manually.



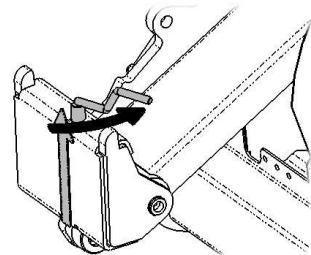
Never move or raise an attachment that has not been locked.

The attachment is mounted to the loader as follows:

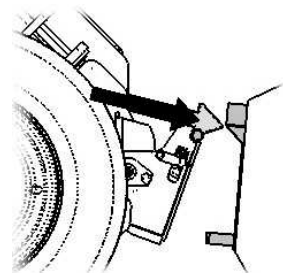
Step 1:



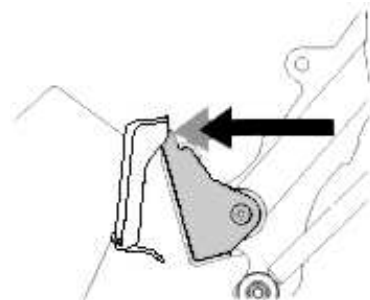
- Lift the quick attach plate locking pin / locking pins up and turn them backwards into the slot so that they are locked in the upper position.
- If your 400-700 series loader is equipped with a hydraulic attachment locking system, see additional instructions about the use of the locking system from the relevant manuals.
- Ensure that the hydraulic hoses are not in the way during installation.



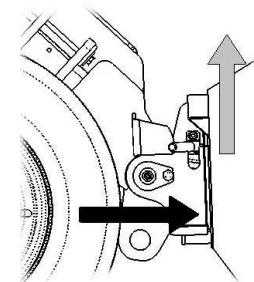
Step 2:



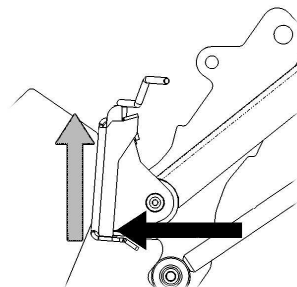
- Turn the quick attach plate hydraulically to an obliquely forward position.
- Drive the loader onto the attachment. If your loader is equipped with a telescopic boom, you can utilise this.



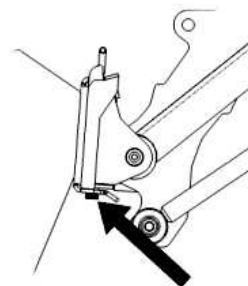
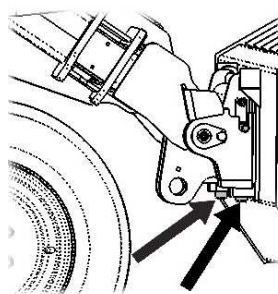
Step 3:



- Lift the boom slightly – pull the boom control lever backward to raise the attachment off the ground.
- Turn the boom control lever left to turn the bottom section of the quick attach plate onto the attachment.
- Lock the locking pins manually as you connect the hydraulic fittings.



Before moving or raising the attachment, ensure that the locking pins are in the lower position and come through the fasteners on both sides. Unlocked attachment may tip on the boom or operator, and cause injuries.



Do not use the automatic return of the locking pins with the log cutter/splitter. Because the quick attach plate of the machine is at high level, tilting of an unlocked log cutter/splitter increases the risk of toppling over. We recommend that you lower the log splitter onto the ground and secure the locking manually.

5.2. Connecting and disconnecting the hydraulic couplings

The attachment is equipped with either standard quick couplings or the multiconnector system. If you want to change the type of the hydraulic couplings for the 300-700 series loaders, contact your Avant dealer or service point for instructions or installation services. The log cutter/splitter A21262 for 200-series loaders is always equipped with standard hydraulic quick couplings.



Never connect or disconnect quick couplings or other hydraulic components while the system is under pressure or if the control lever of the auxiliary hydraulics control lever is locked on. Connecting or disconnecting the hydraulic couplings while the system is pressurized may lead ejection of high-pressure fluid, which can cause serious injuries.

Keep the fittings as clean as possible; use the protective caps for the quick couplings on both the attachment and the loader. Dirt, ice, etc. may make using the fittings significantly more difficult. Never leave the hoses hanging on the ground; place them on top of the equipment.

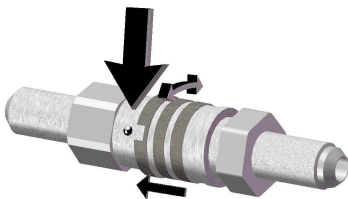
Connecting standard quick couplings:

Before connecting or disconnecting the standard quick couplings, shut down the loader engine and move the auxiliary hydraulics lever on the loader to its extreme positions a few times and also move the control levers of the attachment. This way, there will be no residual pressure in the system and it is easier to connect the fittings.

To connect and disconnect the fittings, move the collar at the end of the female fitting. The hoses should be connected so that the male fitting of the attachment is connected on the loader to the fitting with coloured cap. The second hose is connected on the single male fitting of the loader. Note that the protective caps on the loader and the attachment can be fastened to each other during operation to reduce the accumulation of dirt.



In a 200-series machine, the hoses can be connected by pushing the fitting in the counterpart. When connecting and disconnecting the 200-series hydraulics quick couplings, you should pay attention to their locking. The locking prevents the female and male fittings from being disconnected from each other by accident when the attachment moves. The fitting collar may also turn by accident into the locking position, making it impossible to connect or disconnect the fittings.

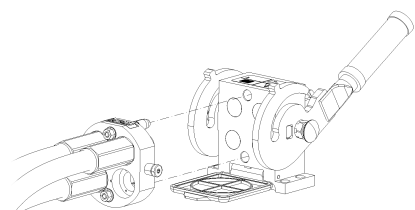


In the 200-series equipment, there is a small groove in the collar of all female fittings and a small ball below the collar. When the fittings need to be disconnected, the above-mentioned groove and ball need to be aligned. Then, the grooved collar of the female fitting is pulled downwards to disconnect the fittings from each other.

Connecting the multiconnector system:

Align the pins of the attachment connector with corresponding holes of the loader connector. The multiconnector will not connect if the attachment connector is upside down.

Connect and lock the multiconnector by turning the lever towards the loader. The lever should move easily all the way to its locking position. If the lever does not slide smoothly, check the alignment and position of the connector and clean the connectors. Shut down the loader and release the residual hydraulic pressure.



Disconnecting the hydraulics:

Before disconnecting the fittings, lower the attachment on solid and level surface. Turn the control levers of the attachment to neutral position and also turn the auxiliary hydraulics control lever of the loader to its neutral position.



When removing the attachment, always disconnect the hydraulic fittings before detaching the quick attach plate, to prevent hose damage and any oil spills. Reinstall the protective caps on the fittings to prevent impurities from entering the hydraulic system.

Releasing residual hydraulic pressure:

In case residual pressure is left in the hydraulic system of the attachment, it is often possible to disconnect the hydraulic couplings, but it may be difficult to connect them the next time. If the fittings will not connect, the residual pressure must be released by turning the control levers of the attachment and the auxiliary hydraulics control lever of the loader, when the engine is turned off.

6. Operation

Once the log cutter/splitter has been mounted according to the instructions and the hydraulic hoses are connected, it is ready for work. Read the original operator's manual of the equipment and also read the following instructions about work site, adjustments and use of the log splitter.



Never perform any adjustments, maintenance or cleaning of the equipment if the splitter can start by accident. Shut down the loader engine first.

6.1. Setting up the log cutter/splitter

- Transport the log cutter/splitter carefully and note the effect of weight at the front to the stability of the loader.
- Choose a level working site, which can carry the load of the equipment and the logs.
- The operating area must be even and it must not be slippery. Clear the area from any obstacles that could cause a tripping hazard.
- Operate the log cutter/splitter in daylight or good artificial light and in well ventilated areas.
- Check regularly to ensure there are no hydraulic leaks. Refer to original instruction manual for maintenance instructions.
- Ensure that the loader and the attachment are used in a safe and intended manner. Smoking near the operating area is prohibited. Do not allow children to operate the equipment.
- Make sure to lower the attachment securely on the ground. Ensure that the parking brake of the loader is engaged and that the splitter can not move during use.



6.2. Use



When operating the log splitter, the loader must be equipped with a hydraulic oil cooler. Without the cooler hydraulic oil may overheat. Overheating of the oil decreases the power and affects the service life of the hydraulic components of both the attachment and the loader, and may also lead to more serious damage to the hydraulic systems. Regularly clean the loader oil cooler core; refer to the loader manual for more instructions.

To start the log cutter/splitter, turn the auxiliary hydraulics lever on the loader to its locked position and set moderate engine rpm, depending on the working pace. Depending on the loader model, two hydraulic pumps may be used to reduce engine noise. Do not exceed the maximum permissible input of hydraulic oil, full rpm must not be used when operating with 600 or 700 series loaders. Note that unnecessarily high engine rotating speed will not improve the splitting force. Operate the log cutter/splitter and perform maintenance procedures as instructed in the original manual of the equipment.



Check the operation and the adjustments of the splitter with a few empty cycles before loading a log.

The lubrication of the log cutter chain saw consumes a small amount of hydraulic oil from the loader. Check the hydraulic oil level regularly.



The log splitter is intended to be used by one operator only.



Never leave the machine unattended with the engine running.



Do not use the attachment for any other purpose than for cutting and splitting firewood.



Read the original instruction manual for any further information of adjustments, use and maintenance of the equipment.

7. Maintenance and warranty

Instructions for maintenance and service are provided with the original instruction manual as well as the spare parts list. Contact your AVANT dealer for parts and service.

The manufacturer of the log cutter/splitter grants the warranty of one year for the log cutter/splitter. Warranty terms are provided with the instruction manual. For warranty claims, contact your AVANT dealer for further information.

The manufacturer of the log cutter/splitter provides the declaration of conformity for the equipment.



AVANT[®]

Instruction manual

JAPA 300



TRH – Hydraulic driven
TR – Tractor driven
E – Electric motor driven
BE – Gas motor driven

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ENGLISH

INSTRUCTION MANUAL

Laitila, Finland, 09/10/2006

FOREWORD

Instruction Manual

The instruction manual is intended as a guidance for an experienced firewood machine operator. Operating the machine requires average level of knowledge and skill.

Right to make changes

All instructions, illustrations and technical information are based on the latest level specifications at the time of the printing of this manual. The manufacturer will, however, carry on with the product development and will therefore retain the right to make changes, without any prior notice, to the design and construction of the machine which will improve the safety and quality of the product.

Machine Identification Plate and contact particulars

In order to ensure fast and efficient service in cases of spare parts orders or in breakdown situations, the Machine Identification Plate information should always be informed to the supplier of the machine or the service technician. Mark down carefully the Identification Plate information on the appropriate space reserved on this page so that the info is always easily available when needed.

Type: Japa Firewood Machine: **300 E5,6T, 300 TR5,6T, 300 TRH4T, 300 TRH5,6T, 300 BE5,6T**

Serial No:

Manufacturing year:

If you are unable to diagnose the fault with the machine, contact the machine supplier who will diagnose the problem together with the manufacturer.

Supplier: Company:.....

Address:.....

Tel:.....

Manufacturer's contact particulars:

Address: Laitilan Rautarakenne Oy
FIN-23800 Laitila Finland
www.japa.fi

Tel Int: +358-2-8571 200

Fax Int: +358-2-8571 201

We have manufactured a safe and reliable machine for you. You have the responsibility for its proper maintenance and for the correct use of the machine. This will ensure that the machine will serve you efficiently and reliably for a long period of time.

Familiarize yourself thoroughly with the machine Instruction Manual before the installation of the machine and before commencing the work with it.

Familiarize yourself also with the machine, its controls and emergency stops before commencing to work with it.

Keep the Instruction Manual always with the machine.

SAFETY PRECAUTIONS

Read before the start-up of the machine

These safety precautions are of general nature. All other relevant safety and health regulations, traffic regulations during transport and common laws must be adhered to when handling the machine. Adhering to the rules and regulations will help to prevent accidents. Familiarize yourself carefully with the machine, with its coupling to the power source, its control and monitoring equipment and with the instructions before the installation and use of the firewood machine. **The machine may be operated only by a person who is familiar with it and with the relevant operating instructions of the machine. The operator must be in sound health, and it is prohibited to operate the machine under the influence of alcohol or any drugs.**

Work site

- The operator work position, when the machine is running, is by the blade feed table (see picture A1 at page 146).
- Select a level and firm work site.
- Organize the site properly, keep it clean from unwanted materials.
- Ensure free passages in the working area during the operation.
- Ensure that the working area is not slippery.
- Ensure adequate general lighting of the work site.
- **The Japa firewood machine is a one man operated machine.** Make sure that there are no other people within the working site.

Personal protection and clothing

- Wear appropriate clothing. The clothing must not contain projections or loose parts which could get caught by the moving assemblies of the machine, thus causing accidents. Special clothing and personal protection used for common forestry work is suitable also for firewood machine operations.
- Use personal ear protection when the noise level exceeds 85 dB(A).
- Eye protection or safety goggles prevent eye damages caused by flying chips which may otherwise occur in spite of the appropriate guards.
- Non-slippery safety boots prevent slipping and protect the operator from foot injuries.
- Non-slippery gloves enable tight and firm grip even on a slippery piece of timber.

Emergency stop

The electric motor driven firewood machine is equipped with an emergency stop which, when pressed, will stop all the machine functions. In tractor driven machines the emergency stop is arranged via a cord connection from the firewood machine stop lever on the top of the machine to the tractor engine stop lever, or to the PTO clutch control lever depending on tractor model in question. Make sure that the safety cord is always within the reach of the operator and test the emergency stop system before starting to work with the machine. The safety cord must be aligned in such a way that there is a clear sight all along the way of the cord. Always, when there has been an emergency stop situation, inspect the machine carefully. The U-J shaft drive must be uncoupled for the period of inspection. Before restarting of the machine, the safety cord must be reconnected. **Note that the stopping of the splitter by the stop lever does not stop the rotation of the chainsaw.**

Hydraulics

Check, with engine running, the functioning of the hydraulic and stopping systems before commencing the work. When the firewood machine is running, there is a high pressure in the hydraulic system. **Leaking high pressure hydraulic oil can penetrate the skin and cause serious injuries.** There is a risk of an injury even when tracing leaks in the system. In case of an injury, contact a doctor immediately (danger of inflammation). Be careful with all hydraulic components. Use proper protections and accessories. Check the hydraulic hoses on regular intervals and renew them if they are worn out or damaged.

INSTALLATION INSTRUCTIONS

The firewood machine should be placed by the log pile to be sawn and chopped so that passages and the working site by the machine are free. Check before starting the machine that all the fastening screws are tight and that all the guards are in their proper places. Before commencing the work, check also that the machine is firmly located. Clean the chainsaw, check its condition and ensure it is sound before starting the machine. Check also the condition of the control and safety equipment before the start-up, and if there are faults, rectify them before you start the machine.

U-J SHAFT DRIVE:

In U-J shaft driven installation the firewood machine is first fastened to the tractor 3-point linkage. It is then centred and locked with the linkage stabilizer bars. The length of the UJ shaft must be checked. Note also that machine end of U-J shaft protective guard must extend at least 50 mm over the shaft (Attachment: U-J shaft instruction E3806-3). A non-contact check must be carried out when raising the 3-point linkage to its up most position. Shorten the U-J shaft if necessary. The speed of the U-J shaft must not exceed 400 r.p.m. For safety reasons this speed must not be exceeded. When the U-J shaft is not coupled to the tractor, it can be lifted onto holding bracket by the machine side.

Tractor:

- The machine drive input is coupled to the tractor PTO shaft by means of a U-J shaft, and the machine itself is supported by the tractor 3-point linkage. The U-J shaft can only be coupled when the tractor PTO clutch is disengaged. Never allow the uncoupling of the U-J shaft relying only on the disengagement of the PTO clutch.
- When starting the tractor engine, the PTO shaft must be disengaged.
- Before engaging the PTO shaft clutch, make sure that there is nobody within the danger zone.
- Always use the correct PTO speed. The PTO shaft must be disengaged when not in use.
- U-J shaft:
- Before engaging the PTO shaft and during the PTO use, make sure that there is nobody within the danger zone.
- Never use anything else except a fault free, protected and approved U-J shaft.
- If a U-J shaft needs to be shortened, follow the manufacturer's instructions.
- In an installed U-J shaft, the locking pin must be inserted and locked in the locking groove on the driven shaft. Check that the shaft is securely in its proper position and locked.
- Connect the shaft guard security chain so that the guard cannot rotate.
- For the uncoupled U-J shaft there is support bracket by the machine side. Never hang the shaft from its security chain.

ELECTRIC MOTOR DRIVE:

- As an electric motor driven version the machine is equipped with a 400 V, 7.5 kW electric motor, which can be connected up a power source by means of a 32 Amp, 5-pin extension cable. The machine is equipped with an emergency stop system. Check that the direction of the rotation is correct.
- If the motor runs the wrong way around, stop it immediately, remove the plug from socket, change the two phase wires around and try again (leave the electric installations to an authorized electrician).

TRANSPORT

The machine is available with lighter (K) and with more massive (R) axles. With K-axle maximum transport speed is 60km/h and with R-axles 80km/h. If the machine is transported on public road it needs to be equipped with lights and 'maximum speed' –sticker. Also other legislations may be set in each country, check the local regulations.

SUPPORT LEGS (B1,B2)

- Take off the holder clamp, lower leg down
- Put the holder back

ADJUSTING THE LENGTH OF LOGS (B3)

- Loosen the fastener of stopper, choose right length, fasten stopper.

INSTALLATION OF THE SUPPORT EXTENSION (C1,C2)

- Take off the holder, lower leg down
- Put the holder back

The rolls guide the log. The roll on user's side also prevents the log from falling on the user.

INSTALLATION OF THE SAW GUARD (pic. C3)

- Before starting the machine, check that the saw guard has been installed.
- remember to lock the saw guard turning shaft
- install the fastening screws in front of the saw guard.

THE SPLITTING ZONE GUARD (pic. C4)

- If the splitting zone guard is delivered separately, You must assemble it before starting to work.
- Check before start that saw-axe gets locked when splitting zone guard is open

INSTALLATION THE DISCHARGE CONVEYOR

Before the firewood machine is taken into use, a discharge conveyor must be fitted to the machine.

- Remove the drive shaft of the firewood machine (pic. C5).
- Make sure, that the axe is in it's most upper position (pic. C6).

Assemble discharge conveyor with fastening bolts (D1A). If the conveyor is hydraulic, attach the hydraulic pipes (D1B). Once hydraulic pipes are removed, the couplings must always be covered with plugs. **Never use the machine, if hydraulic pipes are open!**

Discharge conveyor into the operating position

- D2 Conveyor in transport position.
- D3 Remove the support leg and secure it to the other side of the conveyor.
- D4 Put the locking chain of the leg in a more loose position and lower the conveyor down.
- D5 Unlock the clamp in side of conveyor.
- D6 Straighten the conveyor.
- D7 Lift the conveyor to the operating position leaning on the support leg.
- D8 Lock the upper and lower sections of the conveyor with the clamp.
- D9 Remove the belt bracket
- D10, D11 Lift the conveyor to suitable position and tighten the chain. Make sure, that the support leg is steady on it's place. Maximum angle is 45 degrees.
- D12 Flat belt is tensioned by the upper end hex screws. The same screws are used to adjust the belt to run centrally.

Unloading conveyor into the transport position

Perform the above listed operations in the reverse order.

OPERATING THE MACHINE

- E1 The timber to cut is placed on the feed table and it is pushed against the log length limiter. The cutting handle is pushed gently until the log is cut through. After that you have to hold the switch in the same position for a moment, until the log falls in its place in the splitting chute. This will ensure a successful split.

- E2 The cutting handle is then pushed down further, to its lowermost position, where it then triggers the log pusher equipped with hydraulic cylinder. The pusher then presses the log against the splitting knife. Depending on the splitter knife version, the splitting of a piece of log can be done either to two, four or six parts.

If the log length limiter has been adjusted for a long log, but the log to be split is however short (last piece of timber) and the operating lever is returned quickly back to its neutral position, the limiter goes in front of the log. In this case the operating lever must be held to its lowermost position until the log has been splitted.

- F1 The height adjustment of the splitter knife is done by the control lever.

The chopped firewood is moved automatically by the conveyor directly to a transport load, to a pile of ready firewood further away, or directly into sacks. Moving of the chopped firewood further out increases the work safety as it prevents the chopped wood from piling up against the machine.

NOTES RELATED TO THE OPERATION OF THE MACHINE

Always check the amount of hydraulic oil from the dip rod before starting the engine. Lubrication of saw chain consumes hydraulic oil. Check that the saw chain of the saw is sufficiently lubricated. The amount of saw chain lubrication oil must be adjusted if the oil is cold or warm. In cold weather (below 0°C) it is advisable to "run the engine idle" for about 5-10 min before starting to saw. In idle run the saw engine is running and the hydraulic oil circulates in the engine and becomes more flexible. If sawing is started immediately after start up, the thermal relay of the engine may go off. The thermal relay recovers automatically in about 2 minutes and then the engine may be started again.

The machine guards and protections are there for the safety of the operator. Make sure the all the guards are in their respective positions and in order. The machine is not to be operated if any of the safety guards have been removed, or if they are not in order. Take extreme care when coupling the machine on or off the power source. Always, when commencing any service or repair work, stop the machine, turn off the motor, and in case of electric motor drive, pull out the plug from the socket. When leaving the machine, stop it. Always take special care when working with the firewood machine. Do not allow other people to approach the running machine (i.e. one man operated machine). When the machine is running, be especially aware of the rotating blade and moving splitter.

With electric motor driven machine it must be ensured that the direction of the engine (chainsaw) rotation is correct.

If a piece of wood is stuck in the processor or conveyor, stop the machine at once. Before commencing the removal of the wood waste and sawdust, stop the machine, turn off the motor, and in case of electric motor drive, pull out the plug from the socket. Clean the machine only after it has stopped completely.

Keep the machine emergency stop system on stand-by and clean. Do not cut timber in a bunch. Cut only one trunk at a time. Do not saw timber with nails in it. Take special care when cutting twisted or long branched trees. Ensure that the timber is pressed tight against the cutting table and that it is well supported before commencing sawing. Make sure that the timber cannot wedge itself against the blade. Do not attempt to cut too long trunks. A too long piece of timber will press against the blade. The max diameter of a trunk to be cut is 300 mm and the minimum 20 mm.

Be aware of your back, move the timber from side, do not pull the trunk by the blade end. Remember the right grip. Remove the chopped wood from the vicinity of the machine before it starts piling up against it. Service the machine in due course.

Cover the chainsaw when not in use. Be aware of sharp chainsaw. When opening the sawing guard, stop the machine, turn off the motor, and in case of electric motor drive, pull out the plug from the socket. The splitting zone is safeguarded with an opening guard. When the guard is opened, the splitting action taking place is carried out to completion. Do not open the guard until the pusher has completely stopped. The pusher may move when starting the machine and therefore the guard must be closed prior to start. When the splitting zone guard is opened, the operating lever is mechanically locked up to the upper position keeping the sawchain from rotating and preventing the usage of the cutting and splitting functions. Cutting and splitting functions do not work until the splitting zone guard is closed.

Oils and greases can contain additives which in some cases can be harmful when there has been a repeated skin contact. Follow the appropriate safety instructions when handling these substances.

- Always wear protective clothing, use appropriate hand cream or oil resistant gloves when handling oils and greases.
- Spreading of oil mist to the working site or oil spill will cause a slipping danger.
- Avoid skin contact with oils and greases.
- Never use oil or grease for cleaning the hands.
- Change oily clothes as soon as possible.
- No oily rags etc. in the pockets.
- Waste oil must be collected and when discarding of it, the appropriate rules and regulations must be followed.

Never touch the rotating parts.

Test the functions of all the safety equipment and emergency stops.

SERVICE INSTRUCTIONS

1. Service instructions of the saw chain and the chain blade.

Sawing takes place effortlessly when the saw chain is sharpened correctly. The well sharpened saw chain penetrates into the wood with a little of effort. It is very uneconomical to work with a dull or damaged saw chain. Clean and check the saw chain and make sure that there are no splits in the chain joints and that all the chain rivets are sound. Damaged or worn out chain must be replaced with a new one.

Use only special saw chain sharpening files! Remember the correct tension of the saw chain!

The saw chain has the following chain division: $t = 0.325''$ (pic.G1)

For checking of the division, $t = \text{distance between 3 rivets divided by 2}$. (pic.G2)

Constant sharpening angle 30° (degrees)

- The angles must be the same for all the teeth of the saw chain. If angles vary, the saw chain will move unevenly, it will wear down more quickly and it may even break.
- As these requirements can be fulfilled only after sufficient experience, please do use filing guide!
- All the cutting teeth must be of equal length. If the teeth are of uneven lengths, also the tooth heights are uneven and this will cause uneven rotation of the saw chain, which eventually causes splits in the chain.
- All the teeth must be filed according to the shortest tooth. This could be time consuming, and the best way to do the job is to get it done in a workshop by means of an electric sharpening device. (pic.G3)

Filing of the depth regulator

- depth regulator determines the penetration depth into the wood and thus the thickness of the chip. Depth regulation distance becomes smaller along with filing. Check the depth regulation distance by means of a reduction diagram. If required, file with a flat or triangular file.

The distance between the depth regulator and cutting edge is $= 0.65 \text{ mm}$. When cutting coniferous trees, it could be 0.2 mm bigger, except when the outside temperature is below zero degrees centigrade. (pic.G4)

After sharpening:

- Clean the saw chain thoroughly, remove any stuck chips and filing dust and submerge the chain into oil bath.
- When the saw chain is not used while:
- Clean the chain with a brush and store it in oil/paraffin oil bath.

Servicing the blade

Reverse the blade and file from its side, and clean the groove when necessary.

Replacement and tightening of the saw chain

Turn off the motor before performing maintenance work to the saw chain!

- Lower the discharge conveyor to the ground by means of a winch.
- Loosen the hex screws at the front of the saw bar cover and turn the cover open.
- To tighten the saw chain loosen the retaining nuts (pic.G5) of the saw bar and turn the tightening bolt clockwise. (pic.G6)
- If you need to remove the saw chain, turn the tightening bolt (pic.G6) counter-clockwise until the chain becomes loose.
- The chain should be tightened, so that it would not hang loose under the saw bar.
- Finally retighten the retaining nuts of the saw bar. (pic.G5)
- Check the chain tightness regularly.

2. Check and change of hydraulic oil and saw chain lubrication

- The oil grade is e.g. Mobil Flowrex 46
- The amount of oil is checked from the dip rod in the cap
- The minimum oil quantity is 20 l (the dip rod must get wet)
- Emptying of the reservoir takes place by disconnecting the suction hose
- Oil change should take place once a year, as a principle

3. Bearing lubrications

Bearings used are sealed for life and they do not need to be lubricated necessarily. By the end of the running season it is, however, beneficial to add a little lubricant to the bearings. That can extend their service life. If too much lubricant is forced into the bearing, the seal can be damaged.

4. Service and adjustments of the splitter unit. (pic. G7)

Adjustment of stroke

Most important thing is, that the cylinder never touches bottom and cause pressure peaks which begins to heat up the oil. Consequently pump and the gearbox would be damaged, if that happened. To avoid the bottom stroke, position the pusher to stop 5 mm before the end in the inner position.

Adjustment of the triggering counter plates of the valve (pic. G7B)

Loosen two hex nuts (G7C) and move counterplate a little at the time. Find the place, where bottom is touched and move counterplate so, that cylinder stops 10mm before. Make a few test runs.

1. Left counterplate adjusts, how long strike is.
2. Right counterplate adjusts, how deep the pusher is pulled back.
3. Stroke is correct when the pusher end plate is on the same level with the pullback plate. (G8)

5. Adjustment of the splitter valve

Adjustment screws (1G7D) are the joint swing limiters. Adjustment screw settings take place as follows:

1. The joint sleeve linkage (G7A) is released to its end position.
2. The adjustment screw (G7A) is turned out until the tip of the screw does not reach the joint sleeve any more.
3. The screw is turned in until the screw tip pushes the joint sleeve 2 mm forward.
4. The bolt locking nut (G7E) is tightened.
5. Same actions are repeated on the other side.
6. Screws for the joint sleeve must be tightened.

Now the limiters are set correctly and they will stop the joints at turns, and thus the internal linkage of the valve will be spared from the shocks caused by turns.

The valve neutral position is adjusted by the adjustment screw (G7F). When turning the adjustment screw out, the pusher moves backwards. When the screw is turned in, the pusher moves forwards. Adjust the screw out or in until the pusher, when stopped in the middle of its operating phase, does not move in either direction. Grease should be brushed on the shaft pin when necessary.

8. Replacing hydraulic hoses

NOTE WHEN REPLACING HOSES

- Always before disconnecting a hydraulic coupling, make sure that the hose is free of pressure.
- When replacing hoses, check that they are not twisted or that they do not get twisted when the machine is operating.
- Do not over tighten the hose coupling nut.
- Check the position of the hydraulic coupling.
- Always use the correct size hose clamps.
- Tighten the outside threaded (males) couplings before tightening the coupling nuts.
- Do not use locking or sealing compounds.
- Always use correct size spanners.

DISCARD THE OLD HOSE AND HOSE COUPLING IF:

- Hose has loosened from its coupling (discard also the coupling)
- There is split in the hose or the hose leaks (also when the hose gets wet due to an inside seepage).
- Hose surface has worn down to reinforcement layers.
- Hose contains a flat area or has a permanent change of form.
- There are signs of fire or the hose surface is cracked.
- Coupling is corroded, or contains a permanent change of form.
- Coupling leaks.

Press fit couplings can only be used once.

Hose manufacturer does not warrant the hoses for more than five years.

9. Storing instructions

- Clean the machine thoroughly inside and outside. Dirt accumulates moisture which causes rust.
- Washing can be done with pressure washer, but please, avoid directing the water jet straight on the bearing and other parts which can easily be damaged.
- Lubricate all joint with grease.
- Unpainted, unprotected and worn parts should be treated lightly with oil.
- Do not store machine next to artificial fertilizers.
- Store the machine on firm, even ground. The discharge and feed conveyors can be left in transport position.
- Lubricate the chain flange.

10. Discarding the machine

- The owner, persons or company, owning the machine is totally responsible for discarding the machine.
- Hoses, oils, rubber and plastic parts are to be disposed properly according to the directions of the authorities.
- Iron and other metal parts are to be recycled through machine stripper bays to be reused.
- For discarding the product and handling the waste involved, there are national laws and orders in every country which are to be followed.
- Detailed information concerning the waste disposal and stripping can be obtained from local authorities for the environment.

SERVICE SCHEDULE

Service Item	Action	Daily	6 months / 100 hrs	12months / 500 hrs
Gear box oil	Check Change	X	1.X	next X
Hydraulic hoses	Check Change	X as needed		
Hydraulic oil	Check Change	X		X
Oil filters	Change		1. X	next X
Release mechanism	Check Rubric	X as needed		
Emergency stop	Check at start	X		
Conveyors V-belts	Check Tens - Reply	X as needed		
Chainsaw and saw flange	Clean & Sharpen as needed			
Electric equipment	Check	X		
Motor	Check & Clean	X		

Change of hydraulic oil: Open suction pipe for removing drive.

Gear box oil change: Suction removal of oil or removal of the used oil.

WARNING LABELS

There are a number of compulsory and important instruction and warning labels on the machine. By following these strictly, accidents and breakdowns will be avoided. Locations and notifications on these warnings and instructions are presented in the following illustrations.

H1	Identification Plate and CE-mark.	H10	Stop / Back sticker
H2	Maximum allowed PTO speed	-	In position 2. the pusher reverses when the stop lever is turned to the left.
H3	Japa stickers	-	-In position 1. the pusher stops when the stop lever is turned to middle position.
H4	Height adjustment of the splitter knife (splitting to two parts/ splitting to four parts)	H11	Emergency Stop Stopping of tractor engine: pull in the direction of the arrow Electric drive: press the button
H5	U-J shaft direction of rotation Arrow on the blade guard indicating the direction of rotation.	H12	The oil in the hydraulic system.
H6	Wear personal ear and eye protection	H13	Read the operator manual sticker
H7	Only one user!	H14	Saw lubrication regulating valve
H8	On top of the blade guard: BE AWARE OF ROTATING BLADE	H15	Log length metering sticker
H9	Operating lever 1. neutral 2. the saw chain starts 3. the splitting function starts		

TROUBLESHOOTING

TROUBLE	POSSIBLE REASON	CORRECTIVE ACTION
Chain saw makes contact with guard.	Saw shaft not aligned properly	Open the bearing support screws and realign the shaft. Tighten the screws
Splitting does not function	No oil, or too little oil	Stop immediately, add oil
	Handle does not move down enough	Adjust triggering sensitivity
	Valve adjustment has changed	Adjust the triggering to middle position from the valve
	Litter in triggering system	Clean the triggering system
	Oil too cold	Let oil circulate on free circulation for a few minutes
	Hose broken or leaks	Replace the hose
	Triggering system frozen	Clean the machine always after you have finished working
Slow or powerless splitting movement	Oil too cold. No oil or too little oil	Let oil circulate on free circulation for a few minutes. Add oil. Adjust valve to middle position
Log does not split	Incorrect knife position	Adjust knife height
	A big knot at splitting point	Stop the machine, open the safety guard, turn the log, close the guard
	Exceeds the splitter max limit	Max thickness 32 cm
	Oil pressure has fallen down	Check the hydraulic system
Oil heats up too much	Too little oil, cylinder bottoms and oil pressure remains on, oil circulates via pressure relief valve.	Add oil, adjusted triggering system and cylinder stroke and swing. Check the oil pump
Log lifts up when splitting	Twisted or knotty log	Check the pusher stroke
Electric motor does not start	Emergency stop button down	Pull emergency button up
	Blown fuse	Replace fuse
Electric motor rotates the wrong way round	Two phase wires the wrong way round	A job for an electrician: Change the two wires in the plug the right way round.
Conveyor belt runs on one side	Adjustment has changed	Adjust the return roller at the end of the conveyor. Test drive after adjustment
Log gets stuck at the splitting knife	Wrong stroke of the splitter function. Dull blade	Extend the stroke. Sharpen the knife
Emergency stop does not function	Safety cord not tight or disconnected	Tighten or reconnect safety cord
Log collides with conveyor belt	Conveyor is too upright	Lower the conveyor slope angle (longer conveyor)
When opening the cage, the triggering handle is not locked	Adjustment has changed. Pull spring broken	Adjust the locking mechanism. Replace the pull spring

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TECHNICAL SPECIFICATION

Firewood machine 300

Weight with conveyor	280-310 kg
Weight of firewood machine	205 kg
Length in transport position	2780 mm
Width in transport position	1030 mm
Height in transport position	2450 mm
Saw flange	13 "
Chainsaw	0,325" - 56 links
Length of log	500 mm
Maximum log diameter	300 mm
Electric motor	4,kW
Pusher pressing force	56 kN
Pusher working movement	19 pc/min
Hydraulic pumps capacity	30 l/min
Hydraulic oil pressure	200 bar
Oil tank capacity	27 l
Warranty period	1 year

Noise level of the machine

Noise pressure equivalent level 89,5 dB (A) +/- 3 dB

Noise effect level 100,5 dB (A) +/- 1,5 dB

Vibration

The weight efficiency value of the accelatioeton force by the operator arms
m/s.

Product line

JAPA 300 E5,6T – Electric motor driven model

JAPA 300 TR5,6T – Tractor driven model (PTO)

JAPA 300 TRH4/5,6T – Hydraulic driven

JAPA 300 BE5,6T – Gas engine

Japa 300 Firewood machine spare parts

Part	Part number	Part name	Pcs
1		Frame	1
2	504500A	Extension roller	1
3	504509A	Support	1
4	20104704	Roller Q60-150mm	2
5	502100A	Pusher	1
6	96060	Cylinder 60x570 (5,6t)	1
6A	94206	Piston rod (5,6t)	1
6B	94208	Box (5,6t)	1
6C	94207	Piston (5,6t)	1
6D	94079	Rod seal (5,6t)	1
7	502013A	Turner stop	2
8	96025	Electric motor, 4 kW / 1500 rpm	1
9	96083	Electric motor flange (LS 252)	1
10		Clutch Q86/P98mm E	1
	96111	Clutch Q86/P86mm BE	1
11	94118	Clutch rubber Q86 mm	1
12	97056	Handle 124/25	2
13	96099	Hydraulic pump PLP 20/20(E/TR)	1
14	501020A	Axle	1
15	97016	Wheel IK 400x100	2
16	501040A	Support	1
17	505400A	Splitter knife lifting rod	1
18	505600	Splitter knife handle	1
19	94191	Splitter knife 4-part 12 mm	1
	94202	Splitter knife 5-part 12 mm	1
20	501032	Drive shaft	1
21	96169	Drive clutch	1
22	505500A	Log length limiter	1
23	93071	Locking screw M12x40	1
24	506100A	Splitter guard	1
25	502017	Tap of cylinder	1
26	506010A	Blade guard	1
27	98514	Electric starter 4 kW(380V)	1
28	506018A	Blade guard shaft	1
29	508100C	Saw shaft	1
30	20008003	Saw shaft operating lever	1
31	96095	Chainsaw motor PLM 20.4 S	1
31A	94113	Chainsaw key	1
31B	94139	Chainsaw seal PLM 20/5	1
31C	94136	Support plate to seal	1
31D	94115	Lock ring	1
32	20008111	Drive pulley Z=9	1
33	94125	Guide plate for drive pulley Q50/17	1
34	94114	Lock nut for drive pulley	1
35	96040	Saw flange 13" MLB	1
36	96043	Chainsaw 13" 56	1
37	20104707	Hinge pin	1
38	20008110F	Chain lubrication plate	1
39	96149	Bearing UCFL 204	2
40	93083	Eye screw M8x40	1
41	96015	Springi 4x30x180	1
42	508111	Clamping bar for spring	1
43	96004	Spring 1,25x10x32,3	1
44	507200B	Release stop	1
45	506400	Lifting triangle (only tractor-driven)	1
46	96085	Gear (only tractor-driven) MP2/M	1
46A	94131	Gear sleeve (only tractor-driven) MP2/M	1
47	96086	Hydraulic oil filter MPF 100-2	1
48	94134	Filter element 10/20 MF 100-2	1
49	95142	Oil tank filling neck and dipstick	1

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Part	Part number	Part name	Pcs
50	96109	Valve SD5/1-D	1
50A	94101	U-joint SD5/SD4	2
50B	94096	Spool SD5	2
50C	94100	Protective case SD4/SD4	2
50D	94102	Spool shaft O-ring SD5/SD4	4
50E	94093	Chain flange clamp SD5/SD4	2
51	96106	Valve SD4/1 (180)	1
52	507250A	Stop lever	1
53	507210A	Stop lever bar	1
54	6008111	U-joint sleeve	1
55	6008300	Tappet	1
56	96008	Spring 2x14x100	1
57	6008202	U-joint fork	1
58	504010B	Cover	1
59	95042	Sawchains lubrication angle male union, M8	2
60	95046	Sawchains lubrication hose, 6/4	1
61	95047	Sawchains lubrication controller, 1/8	1
62	501056B	Support	1
64	506302B	Drawbar	1
67	96010	Spring 3,5x16x35	2
68	504011	Leg lug	1
69	504020	Emergency stop (only tractor-driven)	1
70	97129	Rubber of the debris rack	1
71	509602	Fixing iron for the rubber	1
72	507400	Connecting sleeve	1
73	507304	Adjustment piece	1
74	507600	Locking of reversing	1
75	507500	Wobbler	1
76	507700	Safety locking handle	1
77	507302	Intermediate rod of splitting protection	1
78	507010C	Valve frame	1
79	541100A	Axle mounting (only bigger axle)	1
80	97023	Axle (only bigger axle), 750 kg	1
81	97002	Wheel (only bigger axle)145x10,	2
82	97024	Mudguard (only bigger axle)	2
83	20008302	Middle plate	1
84	96098	Hydraulic pump PLP 20/20/8 (only tractor-driven)	1
84.1	96101	Hydraulic pump PLP 20/8/1.5(BE)	1

CONVEYOR

Part	Part number	Part name	Pcs
1A	509000B	Lower frame	1
1B	090000B	Upper frame	1
2	090051	Roller axle	1
3	96136	Bearing, 6205 2rs, per st	2
4	090052	Upper end roller	1
5	96011	Spring 3,5x18x60	2
6A	09018V	Upper side plate, left	1
6B	09018O	Upper side plate, right	1
7	09017	Top end guard	1
8	509025B	Support	1
9	93189	Chain for support, 6mm	1
10	900912832	Belt 3,2m / 6,6m	1
11	9009125A	Belt scraper	7
12	9009126A	Belt joining pieces	2
13	96151	Bearing UCP-207	2
14	509400AHYD R	Lower end roller	1
15	96180	Clutch for shaft Q35/25mm	1
16	50941019	Shaft for motor	1
17	96033	Worm gear 0,18kW	1
18	96074	Hydraulic motor OMP-50	1
19	20009110B	Flange for hydraulic motor	1
20	09225	Belt holder	1
21	97131	Lower end rubber	1
22	509120B	Joining for lower end rubber	1
23	509420AHYDR	Lower end protector (TR&BE)	1
	509420ASÄHKÖ	Lower end protector (E)	1
24	509202A	Flange for electric motor	1

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WARRANTY CERTIFICATE

Type No:

Serial No:.....

This Japa product is covered by a warranty valid for one year from the date of purchase of the product. Warranty covers manufacturing and material faults of the product. Parts and components delivered to replace the faulty parts will not be charged to the customer.

The warranty does not cover:

- * a machine which does not have the original Identification Plate
- * damage caused by abnormal use
- * damages caused by natural wear and tear
- * consumer parts such as V-belts, sawchains etc.
- * wash, cleaning, oil and petrol
- * normal adjustments, operating instructions, service and maintenance

Warranty is not valid during the warranty period if:

- * the machine has not been used or serviced according to the instructions included in the delivery
- * the ownership of the machine has been transferred to a third party

The shipment costs of a machine or a component to be repaired and the labour and travel costs of the mechanic are the responsibility of the owner of the machine. Possible damages which occur during the transport are not covered by the warranty.

This warranty expresses and limits our whole responsibility and it replaces and/or substitutes all other implied or expressed warranties and responsibilities which otherwise may apply.

In a case of a request for reimbursement, based on valid warranty, the Buyer (i.e. the owner) must advise the machine model code, serial number and appropriate details if the purchase receipt, indicating a.o. the date of purchase of the Japa machine in question. The reimbursement request should be presented to our authorized retailer.

All subjects raised under this warranty must be agreed beforehand with the Seller (i.e. the Supplier) or the Manufacturer of the machine before any action is taken.

(Name of the manufacturer)
(Address of the manufacturer)

LAITILAN RAUTARAKENNE OY
FIN - 23800 Laitila Finland
Kusnintie 44
www.japa.fi

(Telephone No of the Manufacturer, Int:)
Fax Int:

+358/ 2/ 8571 200
+358/ 2/ 8571 201

JAPA - PRODUCT

CERTIFICATE OF CONFORMITY

(According to EU conformity certification directive 98 / 37 / EEC - Attachment A II)

(Name of the manufacturer)
(Address of the manufacturer)

LAITILAN RAUTARAKENNE OY
FIN - 23800 Laitila Finland
Kusnintie 44
www.japa.fi

(Telephone No of the Manufacturer, Int:) +358/ 2/ 8571 200
Fax Int: +358/ 2/ 8571 201

It is thus assured that the product to be sold:

JAPA Firewood Machine, model
300 TR5.6T, 300 TRH4T, 300 TRH5.6T, 300 E5.6T, 300 BE5.6T

Optional equipment:.....

Machine serial Number:.....

is in accordance with the machine safety directive 98 / 37 / EEC and the changes made to it, enforced by the decree No 1104/99 of the Finnish Government. The machine is in accordance with the EU directives applicable and in accordance also with the corresponding national regulations:

.....
The harmonized standards and proposals for standards as applicable has been adhered to in the design and construction of the machine:

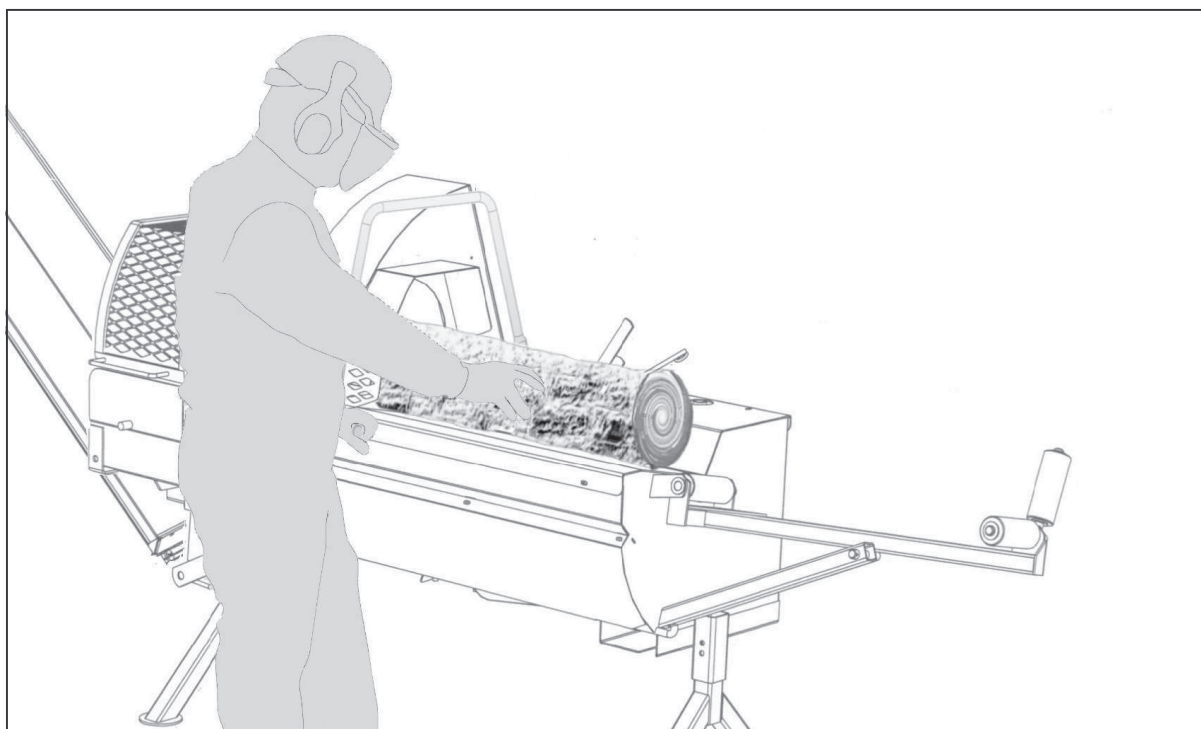
.....
The product described in this certificate has been tested in the usual operating circumstances.

In Laitila.....09/10/2010.....
(date)

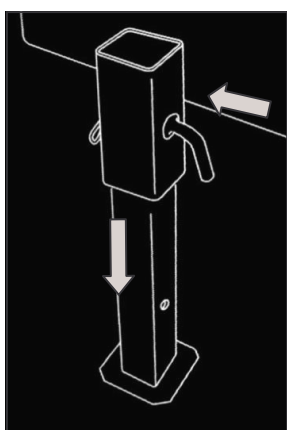
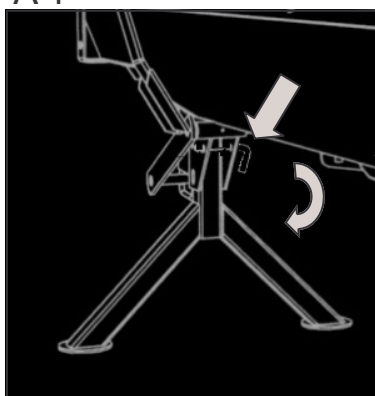


(signature)
Henri Nurminen, Managing Director
(clarification of the name and position)

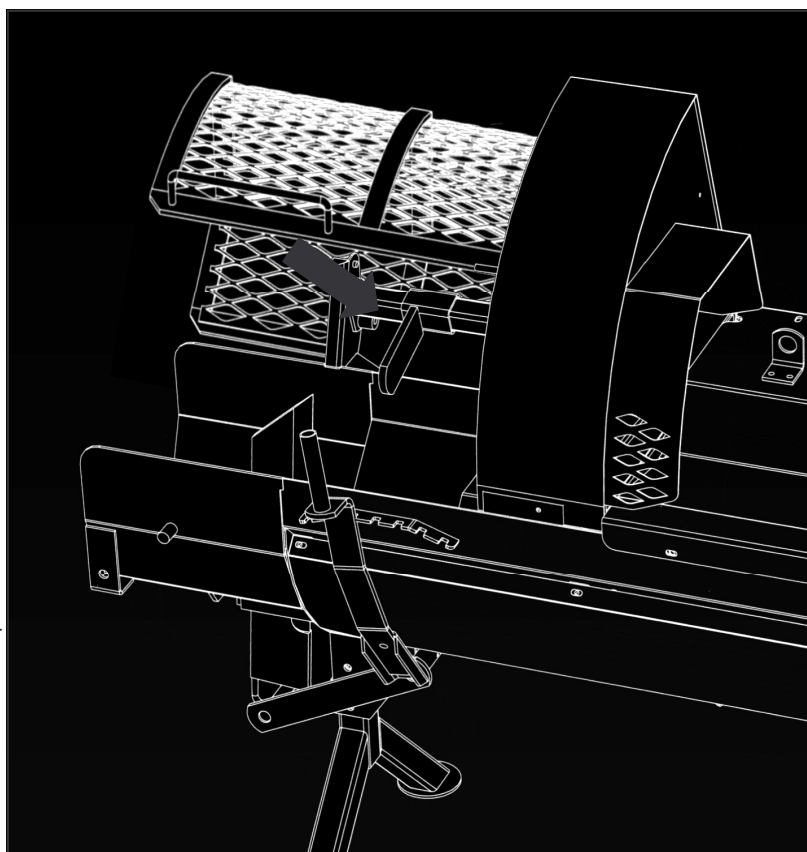
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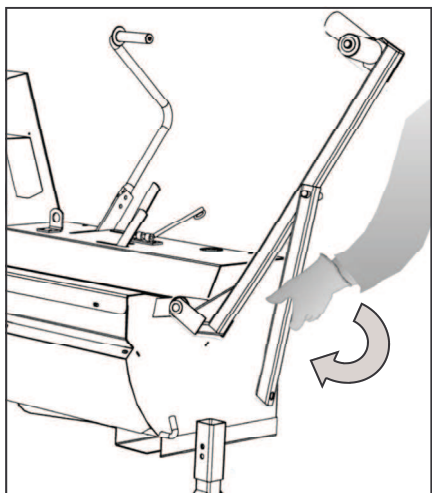


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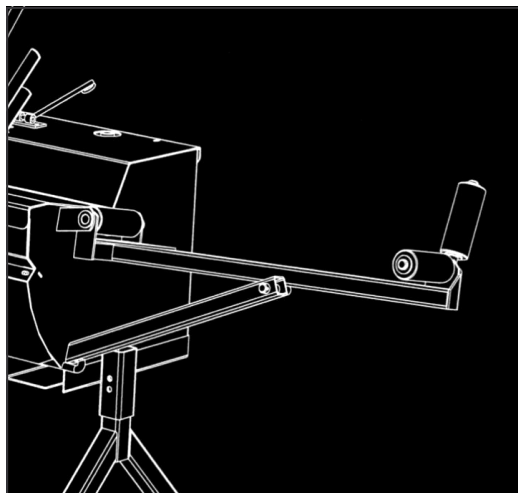


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B3

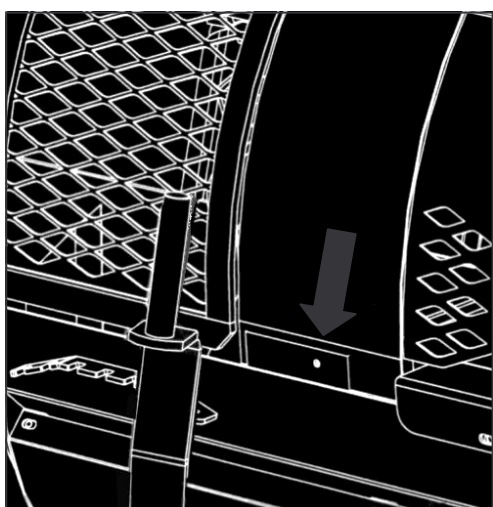




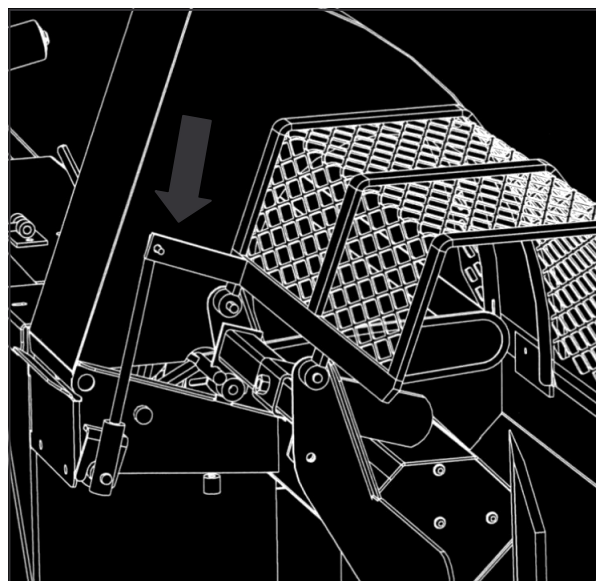
C1



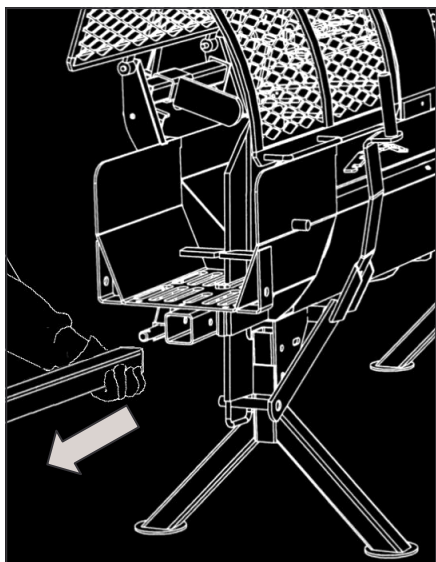
C2



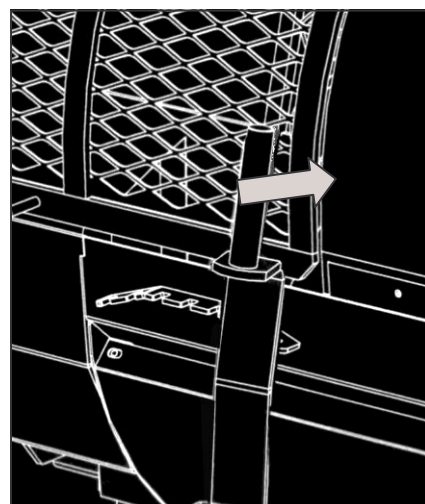
C3



C4



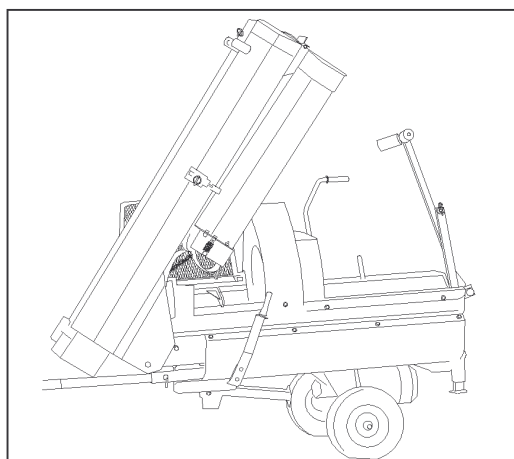
C5



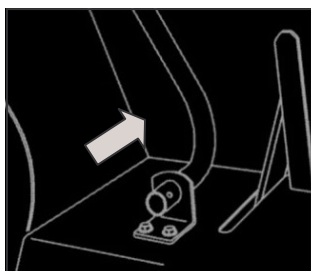
C6



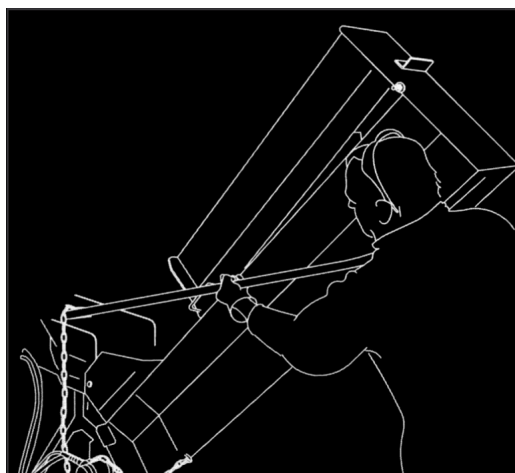
D1



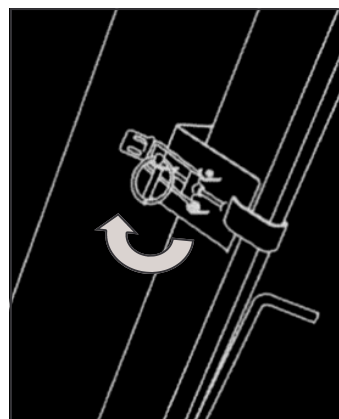
D2



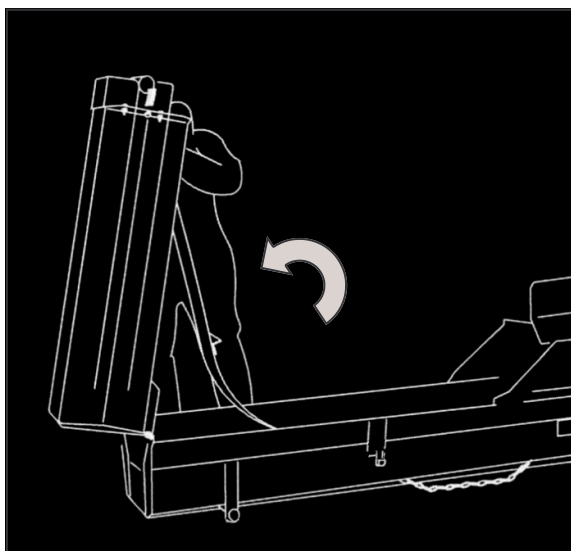
D3



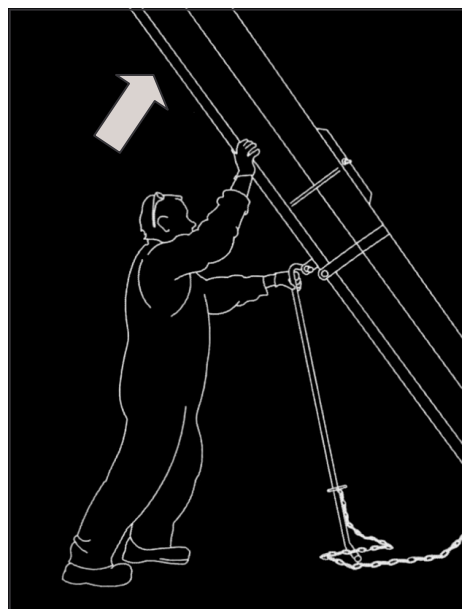
D4



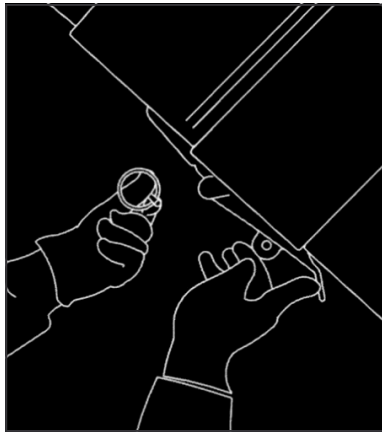
D5



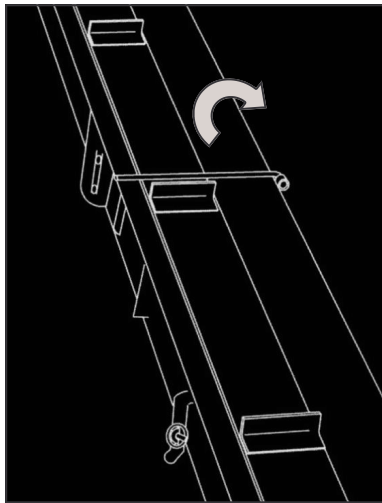
D6



D7

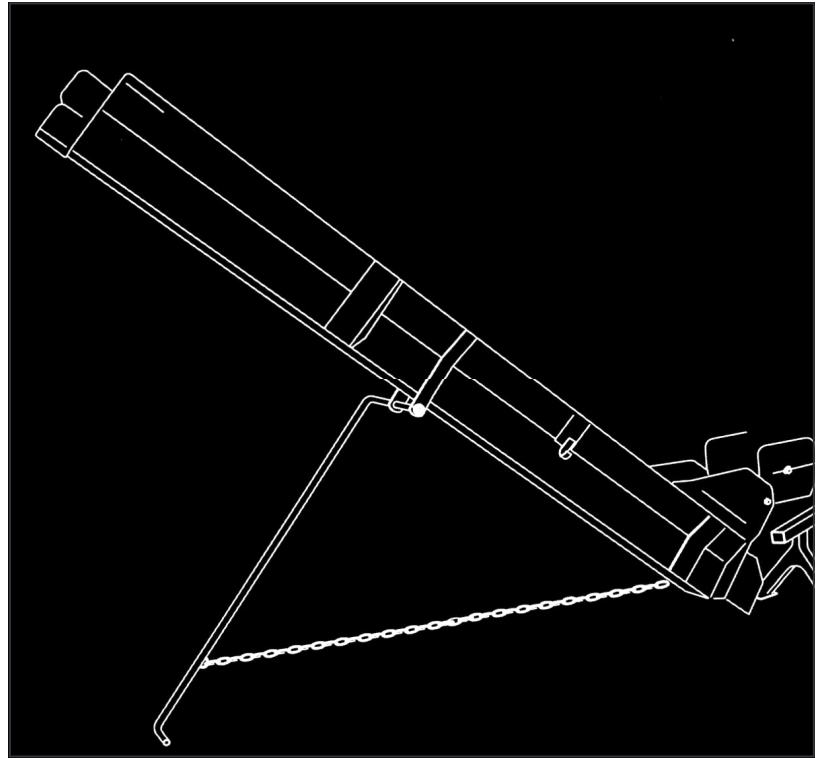


D8

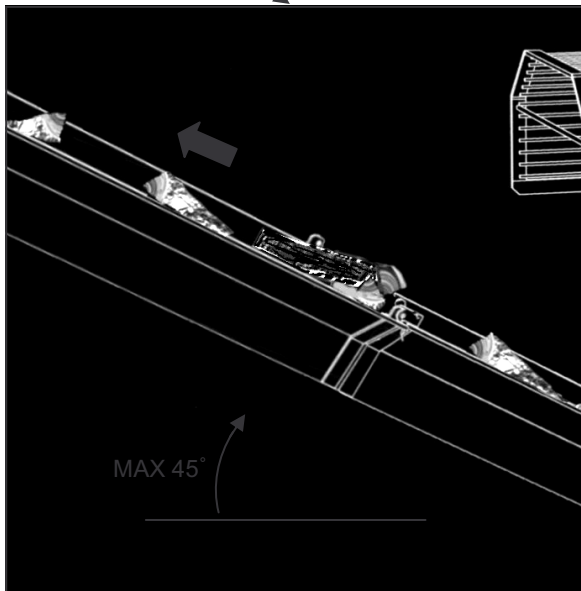


D9

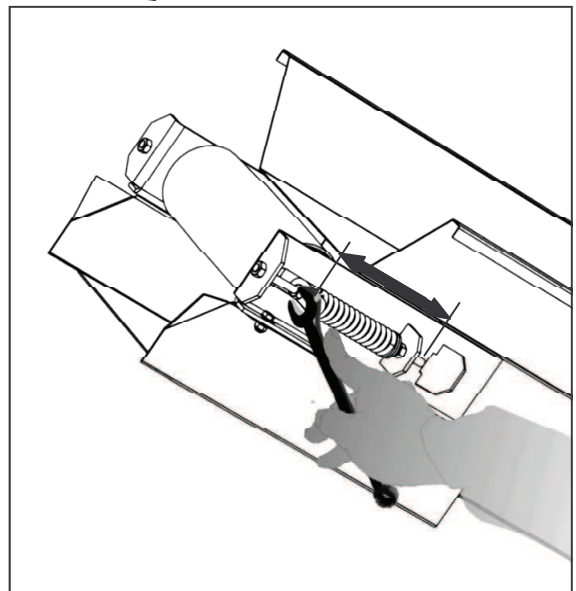
D10

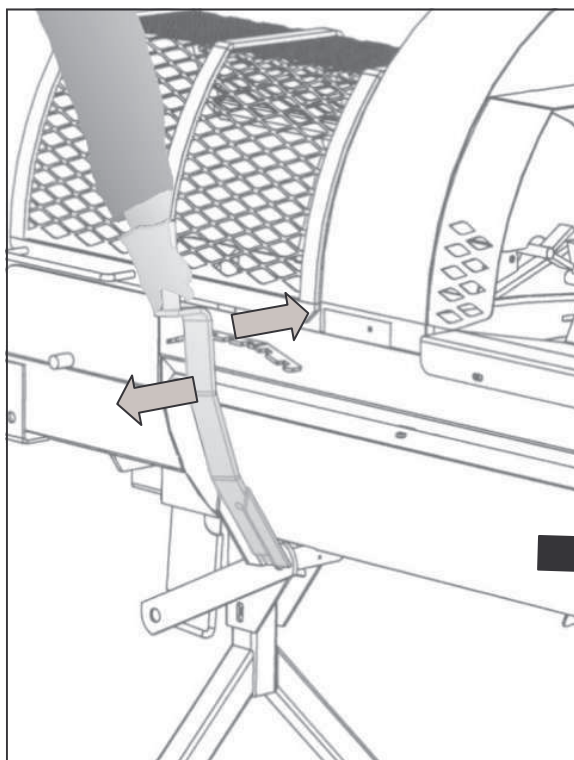
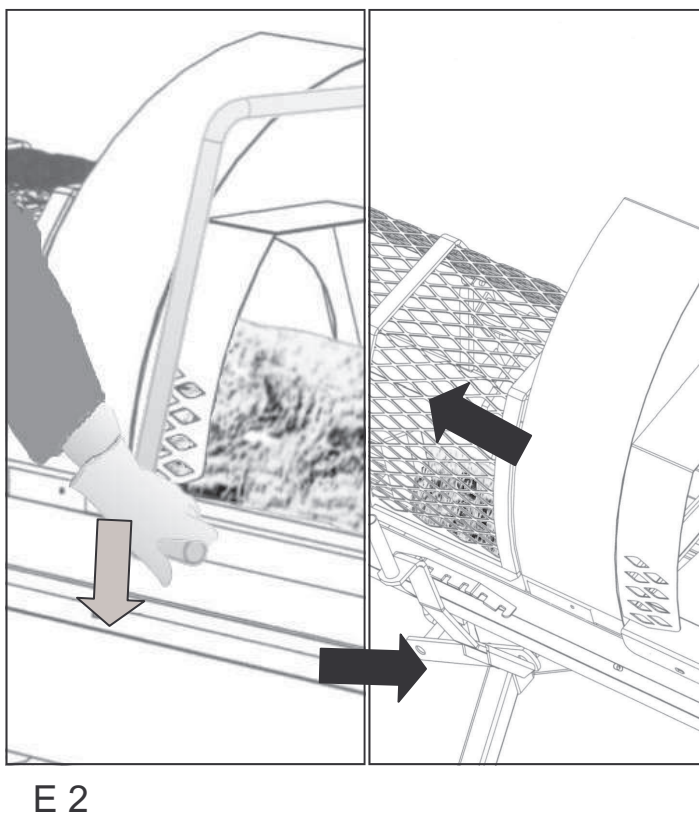
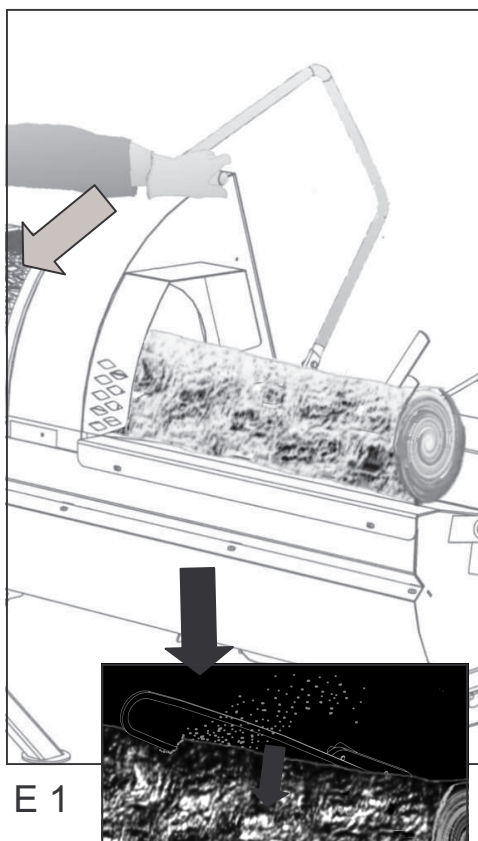


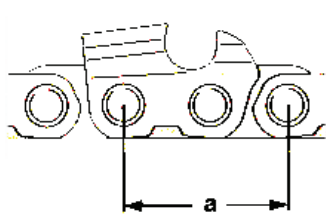
D11



D12

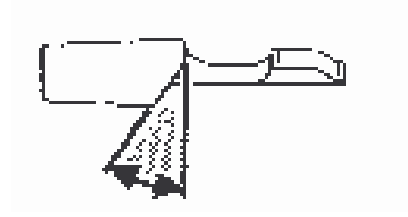






G1

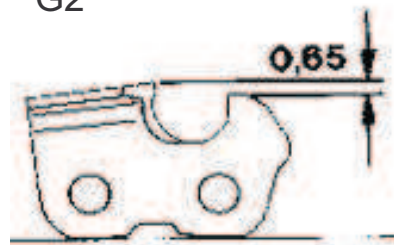
$$t = \frac{a}{2}$$



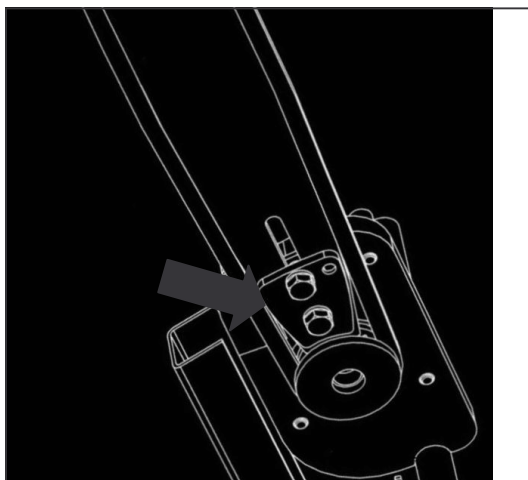
G2



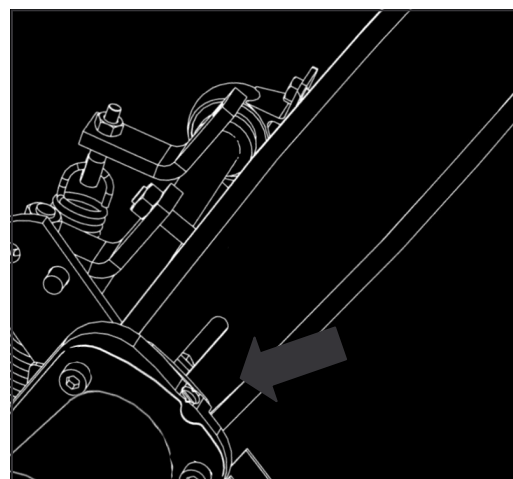
G3



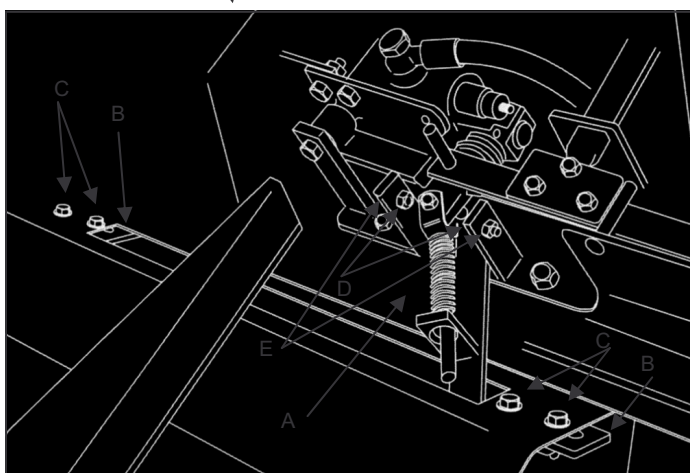
G4



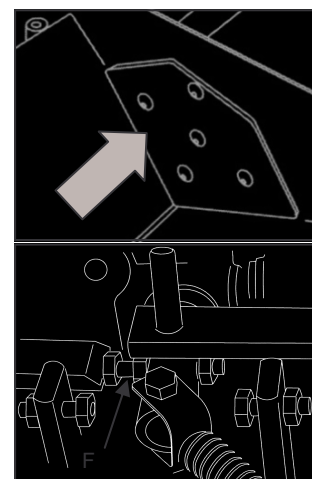
G5



G6



G7 ↓



G8

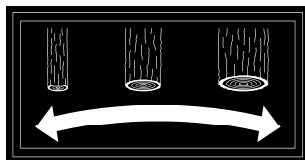
G9



MAX. R. P. M. 400

JAPA

H1



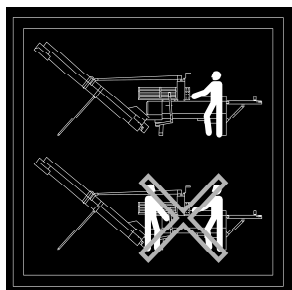
H2



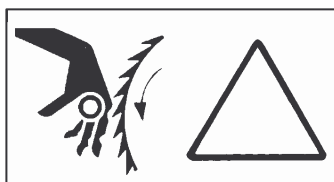
H3



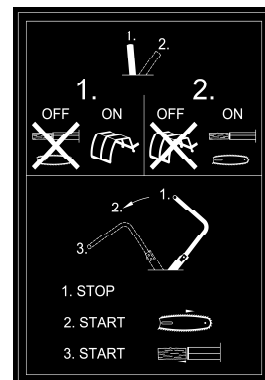
H4



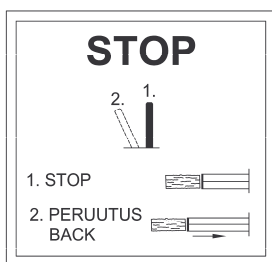
H5



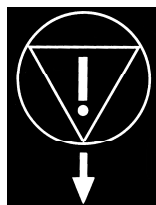
H6



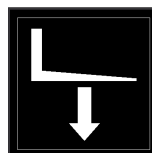
H7



H8



H9



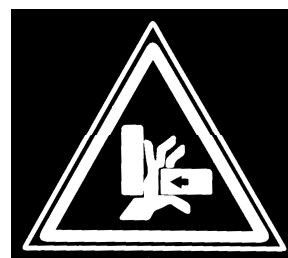
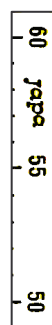
HYDRAULIC OIL
46 S
TEBOIL

H10

H11

H12

H13



H14

H15

H16

H16

